Week 12 Problem Set

Complete Markets and Risk Sharing

Assigned reading: QuantEcon, *Competitive Equilibria with Arrow Securities* (`python.quantecon.org/ge_arrow`
Total: 5 points. Each of the four problems is worth 1.25 points.

Model and notation

Two consumers A, B trade complete one-period Arrow securities. In the infinite-horizon questions, the state is $i \in \{1, 2\}$, $P_{ij} = \Pr(s_{t+1} = j \mid s_t = i)$, Q_{ij} is the date-t price in state i of one unit of date-$(t+1)$ consumption in state j, and $y^k(i)$ is consumer k's endowment. Preferences are common CRRA with discount factor β and risk-aversion coefficient γ. Define $y = y^A + y^B$ and $V = (I - Q)^{-1}$. Use the accompanying notebook for numerical work. Show equations and interpret results, not just numerical output.

Problem 1. A two-period insurance economy (1.25 points)

There are dates $t = 0, 1$ and states H, L at date 1, each with probability $1/2$. Each consumer has utility $\ln c_0 + \frac{1}{2} \ln c_1(H) + \frac{1}{2} \ln c_1(L)$. Endowments are $y^A = (2, 4, 0)$ and $y^B = (2, 0, 4)$, with coordinates $(t = 0, H, L)$.

(a) Define the two Arrow securities and write the date-0 present-value budget constraint for each consumer.

(b) Derive the two Arrow-security pricing conditions from the first-order conditions.

(c) Find equilibrium Arrow prices and all six consumption quantities. Verify goods-market clearing and both budget constraints.

(d) Find each consumer's net Arrow-security portfolio and verify security-market clearing.

(e) Explain why the agents achieve full insurance and why the same conclusion need not hold if aggregate endowment varies across states.

Problem 2. Constant aggregate endowment, infinite horizon (1.25 points)

Let $P = \begin{pmatrix} 0.5 & 0.5 \\ 0.5 & 0.5 \end{pmatrix}$, $\beta = 0.98$, $\gamma = 2$, $y^A = (0.8, 0.2)'$, $y^B = (0.2, 0.8)'$, and initial state $s_0 = 1$. Initial financial wealth is zero.

(a) Derive Q from the aggregate endowment pricing formula and compute state-contingent risk-free gross returns.

(b) Compute V, the two initial consumption shares α_A, α_B, and each consumer's consumption in each state.

(c) Compute $\psi^A = V(\alpha_A y - y^A)$ and ψ^B. Verify $\psi^A(1) = \psi^B(1) = 0$ and $\psi^A + \psi^B = 0$.

(d) Interpret why individual consumption is constant even though individual endowments are not.

Problem 3. Aggregate risk and Arrow prices (1.25 points)

Let

$$P = \begin{pmatrix} 0.8 & 0.2 \\ 0.3 & 0.7 \end{pmatrix}, \qquad y^A = \begin{pmatrix} 0.8 \\ 0.2 \end{pmatrix}, \quad y^B = \begin{pmatrix} 0.4 \\ 0.8 \end{pmatrix}, \quad \beta = 0.96, \quad \gamma = 2, \quad s_0 = 1.$$

Again initial financial wealth is zero.

(a) Calculate aggregate endowment y and the entire Arrow-price matrix Q using $Q_{ij} = \beta P_{ij}[y(j)/y(i)]^{-\gamma}$.

(b) Calculate risk-free gross returns $R_i = 1/\sum_j Q_{ij}$. Which state has the lower return? Explain using marginal utility.

(c) Calculate V, α_A, α_B, and the four state-dependent consumption levels. Verify market clearing and constant consumption shares.

(d) Calculate ψ^A and ψ^B. Verify initial wealth is zero, aggregate financial wealth is zero, and the statewise budget identity $c^A + Q\psi^A = y^A + \psi^A$ (where $Q\psi^A$ is a vector).

(e) Explain what risk is insured and what risk remains.

Problem 4. Interpretation and proof (1.25 points)

(a) Use the Arrow-pricing first-order conditions and common CRRA preferences to show that $c^A(i)/c^B(i)$ is independent of i when the state space is connected and all relevant transitions have positive probability.

(b) Explain the difference between an Arrow–Debreu date-0 contingent claim and a sequence of one-period Arrow claims.

(c) State the condition under which $V = (I - Q)^{-1} = \sum_{t=0}^{\infty} Q^t$ exists as a convergent series. Why is this condition economically useful?